SRM TRP ENGINEERING COLLEGE
Department of Computer Science and Engineering

MINOR PROJECT AS AN ASSIGNMENT

Sub. Code / Title	**CS3492-DATABASE MANAGEMENT SYSTEM**
Class / Semester	**II/IV – CSE A**
Academic year	**2025-26 – EVEN SEMESTER**
Project Title	**Design an application for Virtual and remote guidance counseling system using SQL**
Team Members Register number/Name	**1)814724104301 -AJAY CASTRO M**
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Minor Project Assignment Rubrics

Total Marks – 40

Criteria	Excellent	Good	Fair	Poor	Marks Awarded
Comprehension of the project K2 - (10)	Exceptional understanding of the concept of the project, clearly grasping the core ideas and objectives. (10-8)	Good understanding of the concept of the project, adequately capturing the main ideas and objectives (7-5)	Partial understanding of the concept of the project, with some gaps or limited clarity in grasping the main ideas and objectives. (4-3)	Poor understanding of the concept of the project, lacking clarity and struggling to grasp the main ideas and objectives. (2-1)	
Project Execution K3 (10)	Exceptional execution, demonstrating a thorough understanding of the project requirements and exceptional implementation. (10-8)	Good execution, meeting the project requirements, and demonstrating a solid implementation. (7-5)	Partial execution, partially meeting the project requirements with some noticeable gaps in implementation. (4-3)	Incomplete execution, significant gaps in meeting the project requirements, and limited or no implementation. (2-1)	
Analytical Precision and Validity of Results K4 (4)	Exceptional accuracy in project results, thorough validation, and clear, well-supported conclusions, showcasing exceptional analytical rigor(4)	Good accuracy in project results with thorough validation, leading to well-supported conclusions, reflecting commendable analytical rigor.(3)	Partial accuracy in project results with some validation, though conclusions may lack depth, indicating a satisfactory level of analytical rigor(2)	Lack of project results with insufficient validation, resulting in unsupported conclusions, highlighting inadequate analytical rigor (1)	
Presentation - K3 (8)	Excellent explanation with a thorough understanding, proper eye contact, and clear voice with excellent spoken language (8-7)	Good explanation with a solid understanding, proper eye contact, and Clear voice with good spoken language (6-5)	Limited explanation with fair understanding, poor eye contact, unclear voice, and fair-spoken language. (4-3)	Poor explanation with lack of understanding, no eye contact, and unclear voice with poor spoken language. (2-1)	
Project Report compliance - K3 (4)	Exceptional compliance in terms of content, formatting, reference, and other relevant aspects and adherence to the timeline (4)	Good compliance in terms of content, formatting, referencing, and adherence to the timeline. (3-2)	Partial compliance in terms of content, formatting, referencing, and lack of adherence to the timeline.(1)	Failed to submit the report (0)	
Collaboration and Teamwork - K3 (4)	Excellent collaboration and teamwork, with effective communication, cooperation, and shared responsibilities among team members. (4)	Good collaboration and teamwork, with satisfactory communication, cooperation, and shared responsibilities among team members. (3)	Limited collaboration and teamwork, with some challenges in communication, cooperation, or shared responsibilities among team members. (2)	Poor/No collaboration and teamwork, significant challenges in communication, cooperation, or shared responsibilities among team members. (1)	

DOS:6.4.2026 **Signature Of Faculty**

Create and Maintain a Database for Faculty Profile from Various Departments

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1. Objective

The main objective of this project is to design and implement a centralized database system to store, manage, and maintain faculty profiles from various departments within an institution.

Based on the application's structure and metadata, its primary goals are to:

1. **Manage Academic Records: Provide a comprehensive, full-stack system for administrators to manage faculty profiles, academic roles, and departmental hierarchies with precision.**
2. **Track Departmental Data: Maintain relational database integrity for academic departments, including tracking the number of faculty members and the Head of Department for each unit.**
3. **Streamline Faculty Registration: Offer a seamless interface for onboarding new faculty members, assigning them to specific departments, and recording their designations (e.g., Professor, Assistant Professor, Lecturer) and hire dates.**
4. **Provide High-Level Analytics: Give administrators a dashboard with high-impact statistics, such as total faculty count and the number of academic departments, along with search and filtering capabilities to easily locate specific academic records.**

In short, it is designed to replace manual academic record-keeping with a modern, scalable, and cloud-ready web application (now powered by PostgreSQL and deployed on Vercel).

2)Abstract

Connect Thrive is a digital platform designed to facilitate seamless interaction between users seeking personal, academic, and professional growth and qualified counselors or mentors.The system provides an integrated environment where users can register, explore available services, and schedule virtual counseling or guidance sessions efficiently.

The application incorporates essential features such as user authentication, session booking, real-time communication (chat/video), and feedback management.

Counselors can manage their availability, monitor session requests, and track user progress, while administrators oversee system operations and data management.

The backend of the system is supported by a structured database that ensures secure storage and retrieval of user information, session details, and feedback records.

By leveraging modern web technologies, the platform ensures accessibility, scalability, and user-friendly interaction.The primary objective of Connect Thrive is to bridge the gap between individuals and guidance resources, enabling convenient and effective remote counseling.

The system aims to enhance decision-making, improve user well-being, and promote continuous personal development in a flexible and accessible manner.

3 Problem Statement and Project Execution

In today's fast-paced digital world, individuals especially students and young professionals often face challenges related to academic decisions, career planning, and personal well-being. Access to qualified counselors or mentors is frequently limited due to geographical barriers, time constraints, and lack of organized platforms.

Traditional counseling systems are mostly offline, fragmented, and inefficient, making it difficult to:

- Book sessions conveniently
- Track user progress and feedback
- Ensure continuous communication
- Manage counselor availability effectively

Additionally, there is a lack of centralized systems that integrate user management, session scheduling, and communication in a single platform. This gap leads to missed opportunities for timely guidance and personal growth.

Therefore, there is a need for a reliable, scalable, and user-friendly digital platform that connects users with counselors and provides structured support for continuous development.

4 Project Execution

The **Connect Thrive** project is developed in a structured manner through the following steps:

1. **Requirement Analysis**
   Identify user needs such as registration, session booking, communication, and feedback.
2. **System Design**
   Design the architecture and database (users, counselors, sessions, feedback).
3. **Development**
   Build the application using frontend (HTML/CSS/JS) and backend (Node.js/PHP) with MySQL database.
4. **Testing**
   Test all features to ensure proper functionality and fix errors.
5. **Deployment**
   Host the system on a web platform for user access.
6. **Maintenance**
   Update and improve the system with new features like chatbot and mobile support

GitHub Repository and Project Report Collaboration

GitHub Repository is used to store, manage, and share the project code on a centralized platform. It provides **version control**, which records every modification made to the codebase, allowing developers to track changes and revert to previous stable versions when necessary.

It enables efficient team collaboration, where multiple developers can work simultaneously on different modules of the project using **branching and merging techniques**. Each contributor can independently develop features and later integrate them into the main project without conflicts.

Developers can perform operations such as **push (upload code)**, **pull (download updates)**, and **commit (save changes)** to maintain synchronization across the team. This ensures that everyone is working on the latest version of the project.

GitHub also includes powerful features like **issue tracking**, where bugs, tasks, and improvements can be managed systematically. Additionally, it supports **documentation through README files**, which provide essential information about the project such as objectives, features, installation steps, and usage instructions.

Furthermore, GitHub enhances project development by offering:

- **Backup and security** of code in the cloud
- **Access control** (public/private repositories)
- **Collaboration tools** like pull requests and code reviews
- **Integration with other tools** for testing and deployment

Overall, GitHub ensures better coordination among team members, improves code quality, and supports a structured and efficient development process throughout the project lifecycle.

1) **Repository link:https://github.com/ajay232442/connect-thrive**
2) **Project link: https://id-preview--d41b99ea-0da4-4538-bb10-058afa78c991.lovable.app/**

Project Report Collaboration

Project Report Collaboration is the process in which all team members work together to create, edit, and finalize the project documentation in a structured and efficient manner. It ensures that every part of the project is clearly documented, including the abstract, introduction, problem statement, system design, methodology, implementation, testing, results, and conclusion.

In a collaborative environment, the work is divided among team members based on their roles and expertise. For example, one member may handle system design, another may focus on coding details, while others prepare testing and documentation sections. This division of work improves productivity and ensures that the report is completed within the given time.

Modern collaboration tools such as shared document platforms (Google Docs, MS Word Online) and version control systems like GitHub play a crucial role. These tools allow multiple users to:

- Edit the document simultaneously (real-time collaboration)
- Track changes and maintain version history
- Add comments and suggestions for improvement
- Restore previous versions if any errors occur

Project report collaboration also includes **review and validation processes**, where team members check each other's work to ensure correctness, clarity, and completeness. This peer-review system helps in reducing errors and improving the overall quality of the report.

Additionally, collaboration ensures:

- **Uniform formatting and structure** throughout the report
- **Better communication and coordination** among team members
- **Efficient time management** by parallel task execution
- **High-quality documentation** suitable for evaluation and presentation

Proper collaboration also helps in aligning the report with project objectives and ensures that all important details are included. It results in a professional, well-organized, and easy-to-understand document

References and Future Work

References

The development of the **Connect Thrive** project is supported by various tools, platforms, and official documentation. These resources helped in coding, database design, debugging, and overall system development.

1. GitHub
   Used for version control and collaboration. It allows storing, managing, and sharing project code.
   https://github.com
2. MySQL Documentation
   Used for designing and managing the database structure and executing SQL queries.
   https://dev.mysql.com/doc/
3. Visual Studio Code
   A lightweight and powerful code editor used for writing and debugging code.
   https://code.visualstudio.com/
4. Stack Overflow
   Used for solving programming errors and understanding real-world coding problems.
   https://stackoverflow.com/
5. HTML, CSS, JavaScript Documentation
   Used for frontend design and user interface development.
   https://developer.mozilla.org/
6. Node.js Documentation
   Used for backend development and server-side programming.
   https://nodejs.org/
7. W3Schools Tutorials
   Helpful for learning basic and advanced web development concepts.
   https://www.w3schools.com/

Future Work

The **Connect Thrive** system can be further improved by incorporating advanced technologies and additional features to enhance performance, scalability, and user experience.

1.AI Chatbot Integration

Implement an intelligent chatbot to provide instant responses, guidance, and 24/7 user support using AI techniques.

2. Video Conferencing System

Integrate real-time video communication using WebRTC or similar technologies for better interaction between users and counselors.

3. Mobile Application Development

Develop Android and iOS applications to increase accessibility and allow users to access the platform anytime, anywhere.

4. Advanced Analytics & Reporting

Introduce dashboards to analyze:

- User engagement
- Session effectiveness
- Counselor performance
- System usage statistics

5. Enhanced Security Mechanisms

Implement advanced security features such as:

- Data encryption
- Secure authentication (JWT/OAuth)
- Role-based access control
- Protection against cyber threats

6. Multi-language Support

Add support for multiple languages to make the platform usable for a wider and more diverse audience.

7. Smart Notification System

Provide real-time notifications via email/SMS/app alerts for:

- Session reminders

- Updates
- Announcements

8. Cloud Deployment & Scalability

Deploy the system on cloud platforms (AWS, Azure, or Google Cloud) to:

- Handle large-scale users
- Improve performance and availability
- Ensure data backup and recovery

9. Integration with External Services

Connect with third-party tools such as:

- Payment gateways
- Calendar scheduling systems
- Learning management systems

10. Personalized Recommendation System

Use AI algorithms to suggest:

- Suitable counselors
- Relevant sessions
- Personalized growth plans

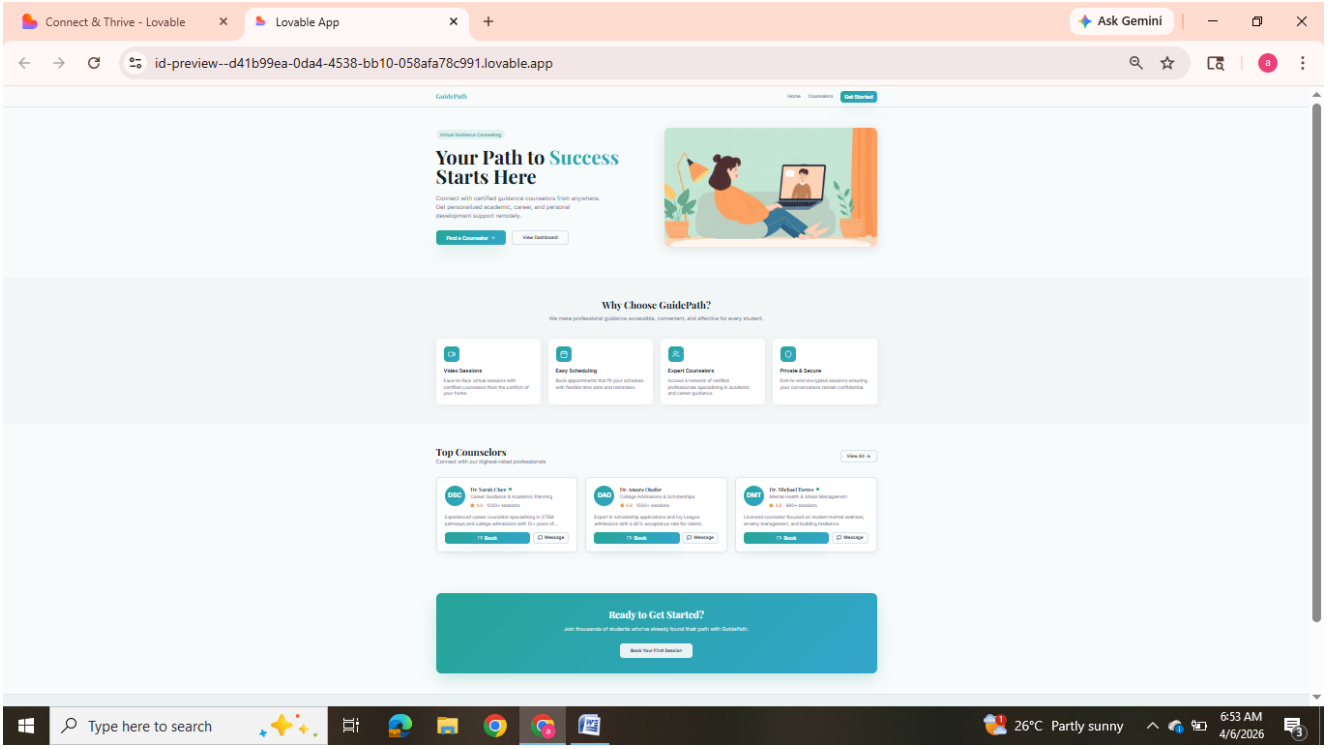
Sample project code:

Index.html:

```
<!doctype html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-
scale=1.0" />

    <!-- TODO: Set the document title to the name of your application -->

    <title>Lovable App</title>

    <meta name="description" content="Lovable Generated Project" />

    <meta name="author" content="Lovable" />



    <!-- TODO: Update og:title to match your application name -->

    <meta property="og:title" content="Lovable App" />

    <meta property="og:description" content="Lovable Generated Project"
/>

    <meta property="og:type" content="website" />

    <meta property="og:image" content="https://lovable.dev/opengraph-
image-p98pqg.png" />



    <meta name="twitter:card" content="summary_large_image" />

    <meta name="twitter:site" content="@Lovable" />
```

```
    <meta name="twitter:image" content="https://lovable.dev/opengraph-image-p98pqg.png" />

  </head>



  <body>

    <div id="root"></div>

    <script type="module" src="/src/main.tsx"></script>

  </body>

</html>
```

Output:


Connect & Thrive - Lovable
Lovable App
Ask Gemini
id-preview--d41b99ea-0da4-4538-bb10-058afa78c991.lovable.app
GuidePath
Home
Counselors
Get Started
Virtual Guidance Counseling
Your Path to Success Starts Here
Connect with certified guidance counselors from anywhere. Get personalized academic, career, and personal development support remotely.
Find a Counselor
View Dashboard
Why Choose GuidePath?
We make professional guidance accessible, convenient, and effective for every student.
Video Sessions
Face-to-face virtual sessions with certified counselors from the comfort of your home.
Easy Scheduling
Book appointments that fit your schedule with flexible time slots and reminders.
Expert Counselors
Access a network of verified professionals specializing in academic and career guidance.
Private & Secure
End-to-end encrypted sessions ensuring your conversations remain confidential.
Top Counselors
Connect with our highest-rated professionals
View All
DSC
Dr. Sarah Chen
Career Guidance & Academic Planning
5.0
1200+ sessions
Experienced career counselor specializing in STEM pathways and college admissions with 15+ years of...
Book
Message
DAO
Dr. Amara Okafor
College Admissions & Scholarships
5.0
1500+ sessions
Expert in scholarship applications and Ivy League admissions with a 98% acceptance rate for clients.
Book
Message
DMT
Dr. Michael Torres
Mental Health & Stress Management
5.0
980+ sessions
Licensed counselor focused on student mental wellness, anxiety management, and building resilience.
Book
Message
Ready to Get Started?
Join thousands of students who've already found their path with GuidePath.
Book Your First Session
Type here to search
26°C Partly sunny
6:53 AM
4/6/2026

Conclusion

The **Connect Thrive** project successfully demonstrates the development of a web-based platform that connects users with counselors and mentors for personal, academic, and professional growth. The system provides essential features such as user registration, session booking, communication, and feedback management in a centralized and user-friendly environment.

By integrating frontend, backend, and database technologies, the project ensures efficient data handling, secure access, and smooth interaction between users and counselors. It overcomes the limitations of traditional counseling systems by offering accessibility, flexibility, and real-time support.

The project also highlights the importance of teamwork through tools like GitHub, which supports collaboration and version control during development.

Overall, Connect Thrive serves as a scalable and effective solution for virtual guidance and counseling. With future enhancements such as AI integration, mobile support, and advanced analytics, the system has the potential to evolve into a more powerful and intelligent platform.